



CSE 6234 Software Design Assignment (20%) TERM 2610

Objectives:

This is an assessment to review your understanding of the fundamental concepts about Software Design.

Course Learning Outcome: Display the qualities of good software design in a prototype software with leadership skills.

Deliverables:

- Working Software Prototype
- Live Presentation
- Presentation Slides - Full Class Diagram Improvised, Component Diagram, Software Architecture
- 10-Minute Group Presentation

Submission Deadline:

Submission deadline for Assignment is **19 June 2026 (Friday), 11.59 pm**

Submission Requirements:

GROUP SUBMISSION

- Working Prototype Software
 - With UI layer
 - Business Logic Layer
 - Database Layer
- Slide Presentation. Content should include:
 - **System Objectives**
 - **System Scope**
 - **System Overview**
 - **Component Level Diagram**
 - **Software Architecture**
 - **Input and Output**
 - **Solution achieved?**
 - **Work Segregation**
- **10-minute group presentation**

GROUP SUBMISSION

- **10-minute Presentation ONLY**

- Slide Presentation on - **Self-Reflection Report:** Each student to share a brief report outlining their contributions, challenges faced, and areas for improvement.

Instructions:

1. Maintain the group combination from the first project. Please work in a group of **4**. No individual submission is allowed. Please maintain your group within the **SAME TUTORIAL** section. *Please update your group member's details in the file provided in Teams/ EbWise.*
2. Develop the software as agreed in the project :

TITLE OPTIONS:

- Ecommerce System
 - AgriTech
 - HealthTech
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3. The final product should be a working prototype system, that demonstrates the effective use of design concepts, design principles, software design patterns to address the specified problem domain.
4. Students have the freedom to define the business rules and determine the assignment scope. The software design should reflect the defined business rules.

Warning:

Be fully aware of assignment deadline date.

Each group is to submit:

1. Working copy of software and all source codes and instruction to your lecturer in eBwise
 2. Group Slide presentation
 3. 5 minutes Live Recorded Presentation
- Group Submission:
 1. 10-minute individual presentation
 2. Slide Presentation explaining contributions- **Self-Reflection Report:** Each student to share a brief report outlining their contributions, challenges faced, and areas for improvement.
 - Late submission, non-contributing members or plagiarism of content is to be penalized with **zero** mark.
 - Individual participation is taken into account by deducting or adding the base marks for the group

Assessment Rubrics:

The OVERALL requirements and marks allocations for Assignment 1 are defined below:

Items	Very High	High	Moderate	Low - 0	Weight
WORKING SYSTEM & – 10% • Functionality & Working Software • Applied Software Design Patterns • User Interface Design • Proposed System Architecture	Excellent on these: • Functionality & Working Software • Applied Software Design Patterns • User Interface Design • Proposed System Architecture	Very Good • Functionality & Working Software • Applied Software Design Patterns • User Interface Design • Proposed System Architecture	• Functionality & Working Software • Applied Software Design Patterns • User Interface Design • Proposed System Architecture	• Functionality & Working Software • Applied Software Design Patterns • User Interface Design • Proposed System Architecture	10
GROUP PRESENTATION & SLIDE PRESENTATION – 10%	Excellent presentation, critically is stated clearly and described comprehensively, delivering all relevant information necessary for full understanding.	Fair and High	Poor	Very Poor	10
INDIVIDUAL CONTRIBUTION AND PRESENTATION – 10% • Software Component involved. • Base contribution • Value-added contribution •	Student demonstrates greater than required knowledge by answering all questions with explanations and elaborations. Clear fair share of wonderful work result	Student is at ease with expected answers to all questions but fails to elaborate. Fair contribution	Group is uncomfortable with information and can answer only basic questions. Poor Contributions	Group does not have a grasp of information, and cannot answer questions about the subject. Very Poor contribution	10
Total marks	20	16	10	5	

SYSTEM & PROTOTYPE RUBRIC

Criteria	Excellent (90-100%)	Good (75-89%)	Satisfactory (50-74%)	Needs Improvement (0-49%)	Weight
Functionality & Technical Execution	Prototype works flawlessly with no bugs or crashes. Includes all key functionalities as per the assignment requirements	Prototype is functional with minor bugs, and most key features are implemented	Prototype has major bugs or missing functionalities, but the basic concept is there	Prototype is incomplete, crashes frequently, or lacks essential features	35%
User Interface (UI) Design	UI is intuitive, well-designed, and aesthetically pleasing. Adheres to user-centered design principles	UI is functional and mostly intuitive but lacks refinement or minor usability issues exist	UI is basic, with some confusing elements or poor design choices	UI is poorly designed, hard to navigate, or not usable	30%
Innovation & Creativity	Shows significant creativity and innovation in addressing the problem. Includes unique features or novel approaches	Some creative elements present, but relies on standard approaches	Limited creativity; prototype largely relies on basic, conventional methods	Lacks creativity or uniqueness; merely repeats standard examples without thought	20%
Code Quality	Code is clean, well-organized, and follows best practices (e.g., appropriate naming conventions, comments, modularity)	Code is readable but has some minor issues with organization or consistency	Code is functional but poorly organized, hard to follow, or lacking documentation	Code is unreadable, disorganized, or contains significant errors	15%
					100%

GROUP PRESENTATION AND SLIDE PRESENTATION

Criteria	Excellent (90-100%)	Good (75-89%)	Satisfactory (50-74%)	Needs Improvement (0-49%)	Weight
Understanding of Design Patterns & Component Level Design	Demonstrates a thorough understanding of multiple design patterns (e.g., Singleton, Factory) and integrates them effectively into the prototype	Shows a good understanding of design patterns, with minor flaws in the integration	Demonstrates basic understanding, but has significant issues in the application of patterns	Shows little to no understanding of relevant design patterns or incorrectly applies them	35%
Presentation & Documentation	Clear and detailed documentation that explains design choices, functionality, and future improvements. Excellent presentation of the prototype	Documentation is clear but lacks depth in some areas. Presentation is good but could be more polished	Minimal documentation with gaps in explaining important aspects. Presentation is unpolished	No or very poor documentation. Prototype is poorly presented	35%
Team Collaboration	Evidence of strong collaboration with clearly defined roles. Team worked together seamlessly	Team collaboration is good but uneven workload distribution or minor issues	Team worked together but encountered some significant challenges	Lack of collaboration; project was completed by one member or poor communication evident	30%
					100%

INDIVIDUAL CONTRIBUTION

Criteria	Excellent (90-100%)	Good (75-89%)	Satisfactory (50-74%)	Needs Improvement (0-49%)	Weight
Participation & Engagement	Actively participates in all project phases, consistently attends group meetings, and contributes valuable ideas	Participates regularly in most project phases, attends most group meetings, and provides useful input	Participates sporadically, attends meetings but does not contribute significantly	Rarely participates, misses meetings, or contributes little to the group effort	25%
Quality of Individual Contributions	High-quality contributions that significantly enhance the project (e.g., code, design elements, or documentation)	Good quality work that contributes positively to the project, but lacks some polish	Work is completed but with notable quality issues, requiring major revisions	Contributions are incomplete, incorrect, or of poor quality	25%
Initiative & Responsibility	Takes initiative, goes beyond assigned tasks, and proactively solves problems without needing to be prompted	Completes assigned tasks well and occasionally takes on additional responsibilities when needed	Completes assigned tasks, but needs reminders or encouragement to take initiative	Frequently avoids responsibility, does not complete assigned tasks, or relies on others to complete the work	25%
Communication Skills	Communicates effectively, keeps the team informed, and actively participates in discussions to move the project forward	Communicates well, though may occasionally miss updates or discussions, still contributes to team coordination	Communicates inconsistently, occasionally misses key updates or discussions, affecting team cohesion	Rarely communicates or fails to provide essential updates, which negatively impacts the team's progress	25%
					100%